

Baseline Choice Template

One baseline per claim, seven roles, and a gate before adding an eighth

1 What a baseline is for

A baseline answers one question: what would you have got without the thing you are claiming credit for? Different baselines answer different versions of that question, and a paper needs the ones matching its claims. That is the whole principle, and everything below is its consequence.

Two failures follow from ignoring it, and they point in opposite directions. A paper with a baseline no claim needs has spent runs on a comparison nobody asked for. A paper with a claim no baseline tests has an assertion sitting where evidence should be, and a reviewer will find it before a reader does.

2 Seven roles

The taxonomy question 21 asks for is not a taxonomy of algorithms. Rainbow and a vanilla Dreamer agent are different kinds of algorithm but they can hold the same role, and one algorithm can hold two. The useful division is by the question the comparison answers.

Role	The question it answers	This paper
Floor	What does no skill at all get in this environment?	Sleep, Random
Operational reference	What does current practice get, without learning anything?	Expert-Inspired Heuristic
Paradigm contrast	Is the gain the family of method, or this method?	Rainbow, IQN
Recipe ancestor	Is the gain the family, or my additions to it?	the no-factoring, no-opponent cell, which is a vanilla Dreamer agent
Component ablation	Which addition, and do the additions interact?	the 2×2 , one cell short
Confound control	Is the gain the extra information rather than the architecture?	absent
Ceiling	What would the same information buy if it were simply handed over?	absent

Three of these are the ones security reviewers ask for by name and machine learning venues often skip. The floor is how a reader knows what the numbers mean at all. The operational reference is what a practitioner compares against, and a learned defender that does not beat the deployed playbook has no case for being deployed. The confound control is the one that decides whether a result is believed.

The recipe ancestor is the role most often filled without being labeled. If the bottom row of an ablation table is the standard agent, saying so converts an ablation into a paradigm comparison at

no cost, and a reviewer who asks for a vanilla baseline is asking for that sentence rather than for a run.

3 The map, one row per claim

```
Claim, as a sentence a number could contradict
Role that tests it
The arm that fills the role
What is held equal between the arms
What is not equal, and is stated instead
Present, missing, or filled by an existing row under another name
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Write the claims first, from the abstract and the contribution list, and only then the arms. Doing it the other way produces a table of everything that was run, which is a different document.

The last field is the one that saves runs. A missing row is often an existing row that has not been named, and naming it is a writing change.

4 Principles

One baseline per claim. A comparison no claim needs is not evidence, it is a run you have to defend.

Match the resource the claim is about. A sample-efficiency claim is quantified in environment interactions and has to report them. A deployment-cost claim is quantified in wall-clock on the target system. The two order methods differently, and holding one fixed does not hold the other fixed.

Match the mechanism, not the information. A baseline that receives the same bits through a different pathway tests something else. Predicting a signal forces the representation to encode structure related to it; observing the same signal as an input feature does not. The two uses are not interchangeable, and a reviewer can say the variant answers a question nobody asked.

Tune the baseline as hard as your own method, and say how. Default settings carried over from another environment are the standard way a comparison goes wrong, and the fix is cheaper than the run it discredits.

Prefer the baseline the field already recognizes. A bespoke variant that loses proves less than a standard one that loses.

In security, the floor and the operational reference are not optional. The systematization of deep reinforcement learning work in this area finds that half the corpus does not separate the contribution of the learned policy from the contribution of how the task was set up, and the cure it names is an evaluation under random action sampling.

Add a baseline only when it removes a confound nothing else removes. Not when it merely adds a comparison.

Weigh the asymmetry before promising a run. A confirming result restates what a corrected sentence already says. A disconfirming result damages the headline claim, and during a shepherded

revision there is no room left to reconsider.

5 Model-based against model-free

The like-for-like question has a clean answer and an inconvenient one.

The clean answer is the list of things that must be identical: the observation stream, the action space, the reward, the episode structure, the seeds, and the effort spent searching for hyperparameters. Anything on that list left unequal is a confound, and the fix is a run rather than a sentence.

The inconvenient answer is that two items can never be made equal, so they get stated instead of matched. The first is the number of gradient updates per environment step, which is what the model-based method is buying with its extra compute; equalizing it would remove the mechanism under study. The second is the auxiliary supervision each method receives. The rule for the second comes from the security systematization and is worth using verbatim: comparisons between approaches should be performed with equivalent information and capabilities.

Which leaves the axis. The resource a model-free agent spends is environment interactions. The resource a model-based agent spends is compute per interaction. A fixed wall-clock budget on a live testbed is a deployment-cost comparison, and it is the right comparison for a paper about training in the operational network. It is not by itself a sample-efficiency comparison, and a paper claiming sample efficiency has to report the interaction count for every arm. That is one number per arm, recoverable from logs, and it answers the reviewer who asked what the episode count actually counts at the same time.

6 The gate for adding one

Three questions, in order, before any new arm enters a response or a revision.

1. Which role in the table above is empty, and which claim is left untested by its emptiness? If no claim is untested, the answer is a sentence naming the existing row, not a run.
2. Would the new arm test the mechanism, or only the information? If only the information, name the mechanism-matched version and offer that one instead.
3. Is there a cheaper measurement that answers the same question from the other direction? A surviving log from an earlier attempt answers a confounding worry at the cost of a search rather than an experiment.

What leaves the gate is one of three things: a commitment naming a configuration and the table it lands in, a decline stating a reason that is never effort and never the word limit, or a numbered offer with what it would show.

7 Filled, for this paper

Claim	Arm	State
Model-based learning beats model-free under a fixed live budget	Rainbow, IQN	present; the budget matched is wall-clock, so the interaction counts are the missing number
The learned defender exceeds current practice	Expert, Random, Sleep	present
The two additions beat the plain recipe	the no-factoring, no-opponent cell	present, and equivalent to a vanilla Dreamer agent; saying so is the whole answer to the comment
Each addition contributes separately	the 2×2	one cell short, and the missing cell is the informative one
The gain is the architecture, not the privileged supervision	none	untested

The last row is the one the comment about baseline fairness points at, and it is worth being precise about what the comment says. It names a writing defect: one section states that the model trains with ground-truth attacker labels, another asserts that no baseline receives privileged information the model does not also see, and the second sentence runs the wrong way to establish equivalent supervision. Replacing that sentence with an account of what each method received during training, and what remains available at deployment, fixes what broke.

Three candidates could fill the row, in increasing order of what they cost and decreasing order of how likely they are to be needed. A search through the early model-free attempts, which answers the confounding worry from the other direction if the logs survived. Rainbow with the same auxiliary prediction head, which matches the mechanism. Rainbow with attacker state appended to its observation vector, which matches the information and not the mechanism, and is therefore the one to decline.

8 Sources

- The claim-to-baseline table for this paper, the reading of what the comment about baseline fairness actually asks for, and the verdict on the appended-feature variant are in `baseline_choice.tex` in this package.
- McFadden et al., *SoK: The Pitfalls of Deep Reinforcement Learning for Cybersecurity*, for the equivalent-information rule, the random-action ablation, and the requirement that applying this family of method be justified against the in-domain alternative rather than only against other agents.
- Patterson, Neumann, M. White and A. White, *Empirical Design in Reinforcement Learning*, JMLR 2024, on calibration baselines, designer bias, and tuning the baseline as hard as your own method.
- Henderson et al., *Deep Reinforcement Learning that Matters*, for the worked case where retuning a baseline reverses the result.

- Arp et al., *Dos and Don'ts of Machine Learning in Computer Security*, USENIX Security 2022, for the inappropriate-baseline pitfall as the security community states it.
- The reviewer comments quoted here are reproduced in the reviews file of this package.